Labour Supply 1: Individual Attachment to the Labour Market

EC306- Wages and Employment

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Goals of This Section

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- Introduce common labour market concepts
- Build the labour supply model
- Derive labour supply curve
- Examine effects of labour market policies within that model

Key Labour Force Concepts

Introduction

- The labour market involves several individual decisions about work
  - It is not simply whether to work or not
- In the news you hear about statistics that quantify these decisions
- In this section we cover the most common ones

Labour Force Participation

- The first decision is whether to make yourself *available* for work
- **Labour Force**: people in the eligible population who participate in labour market activities as either employed or unemployed
  - A measure of the people that could work or are working
- Eligible population includes
  - Population 15 years and over
  - Non-institutionalized
  - People not living on reserve
  - Civilians (non-military)

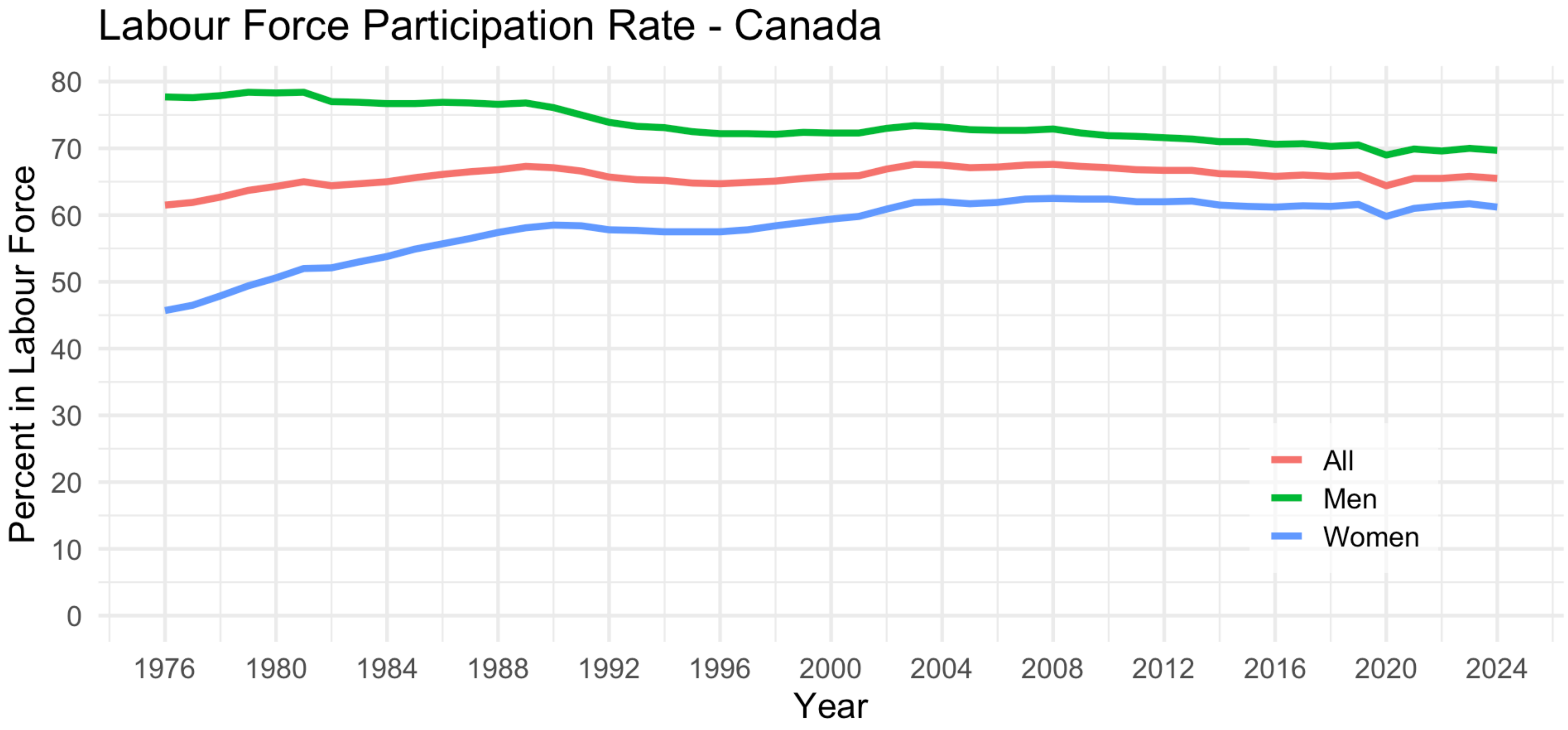
Labour Force Participation

- Eligible people who are not part of the labour force include
  - Retired people
  - People who do only unpaid work in the home
  - Full time students (who are 15 years old and over)
  - People who just give up on looking for work
- **Labour Force Participation Rate (LFPR):** The share of the eligible population (POP) in the labour force (LF)

$$LFPR = LF/POP$$

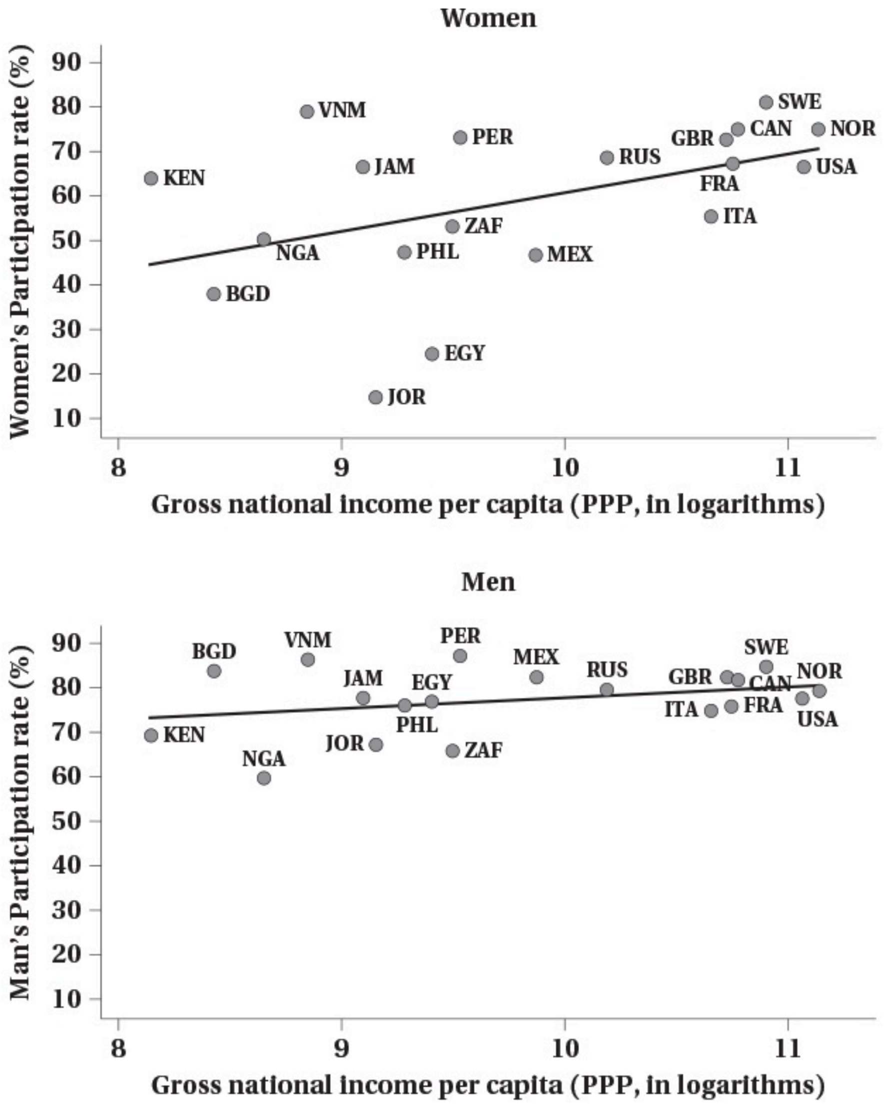
- The LFPR is a common statistic reported in the news

Labour Force Participation



Source: CANSIM Table 14-01-0327-02

Labour Force Participation



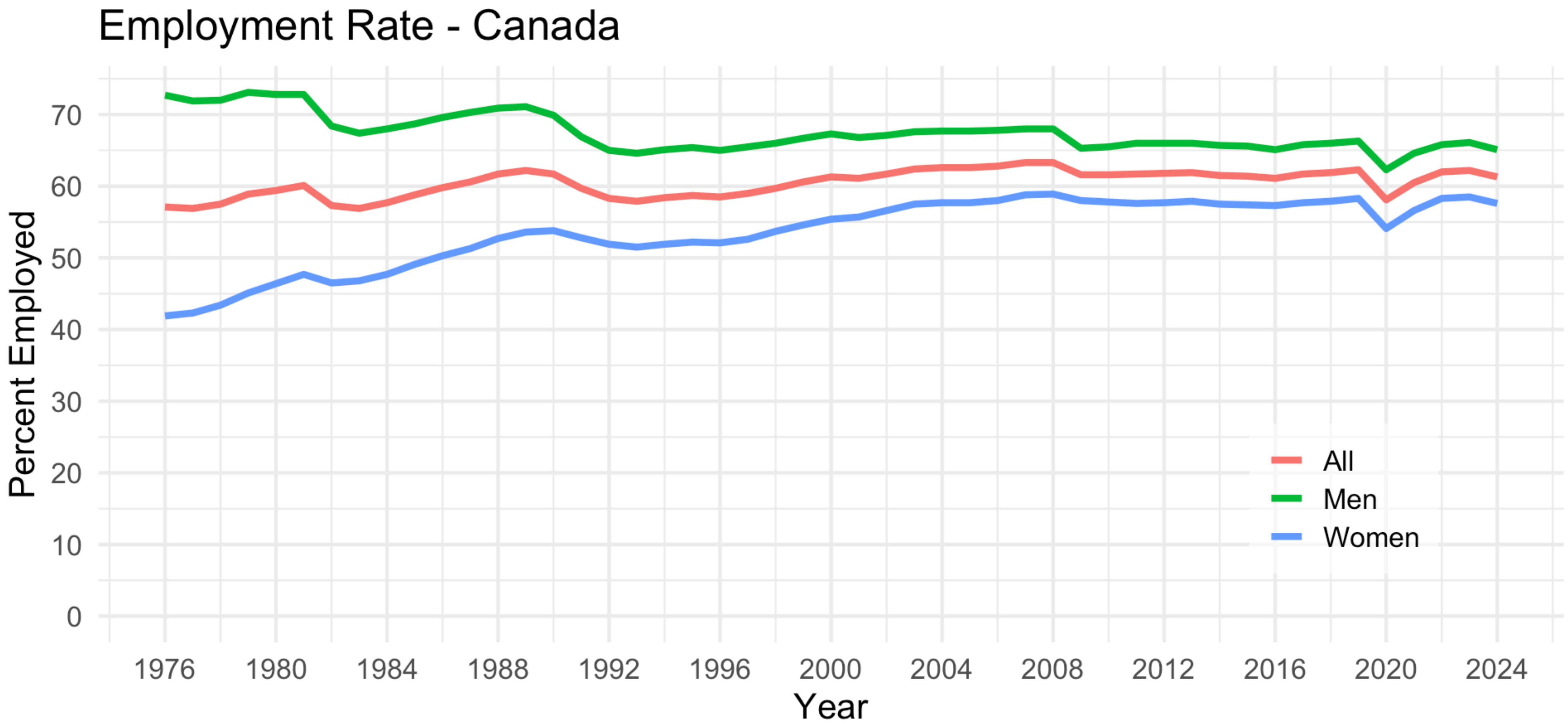
Employment and Unemployment

- People in the labour force are either employed or unemployed
- **Employed (E)**: people who worked during the reference period
  - Includes people normally employed but temporarily absent
- **Unemployed (U)**: people who did not work during the reference period but
  - were available for work and actively looked for work in the past four weeks
  - were waiting to start a new job within the next four weeks
  - on temporary layoff and expecting to be recalled to that job
- The employment rate (ER) and unemployment rate (UR) are

$$ER = E/POP$$

$$UR = U/LF$$

Employment and Unemployment



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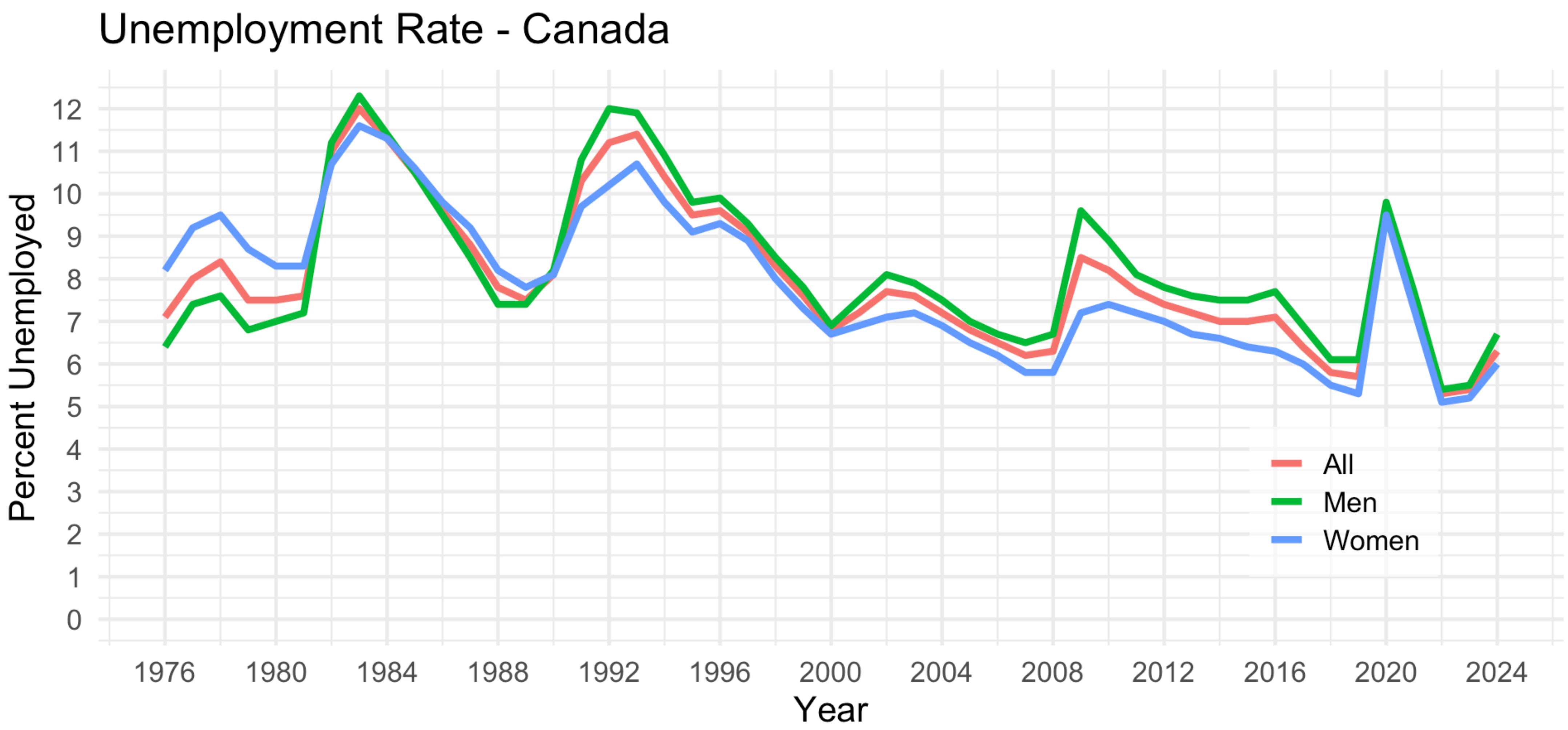
Employment and Unemployment



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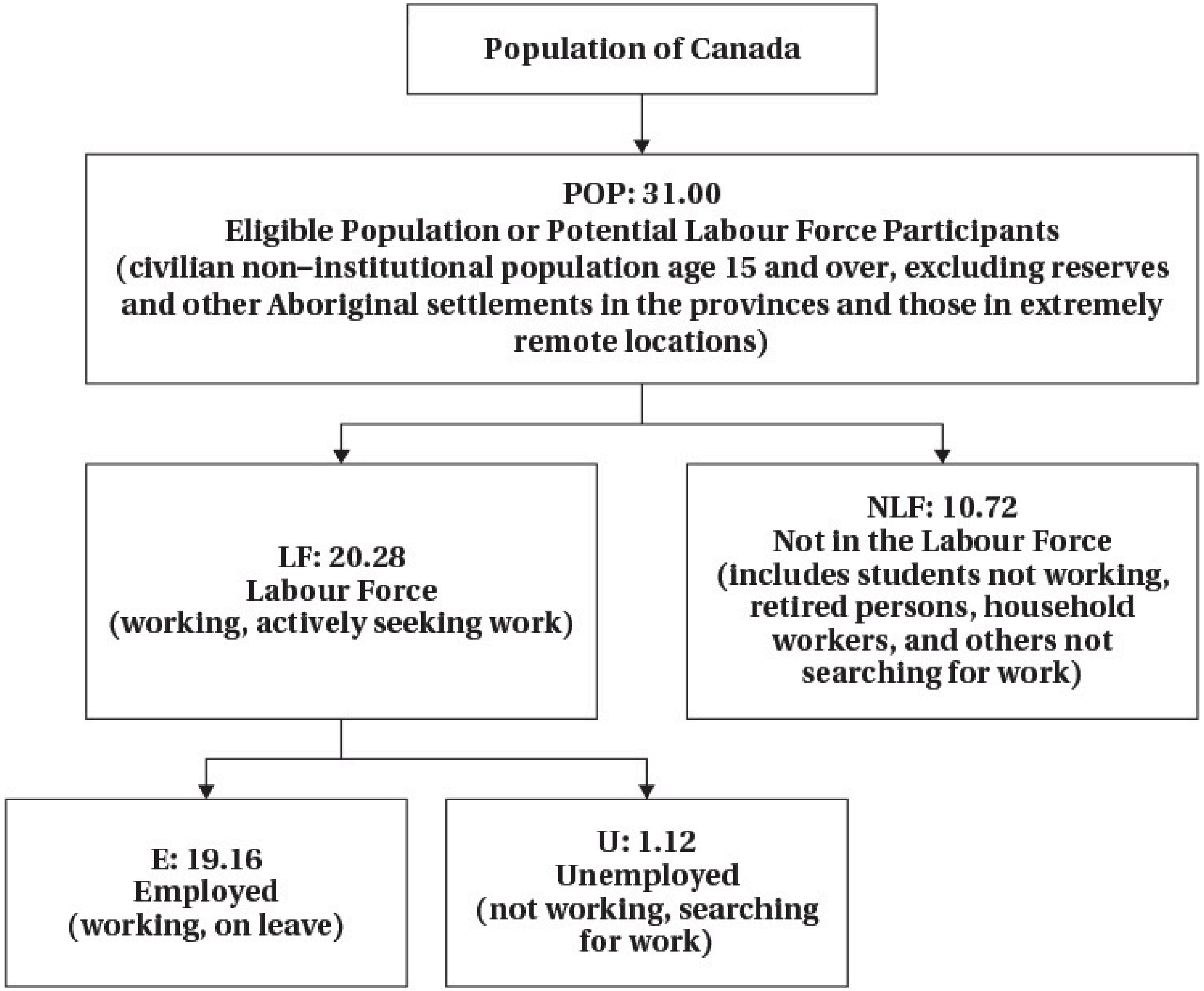
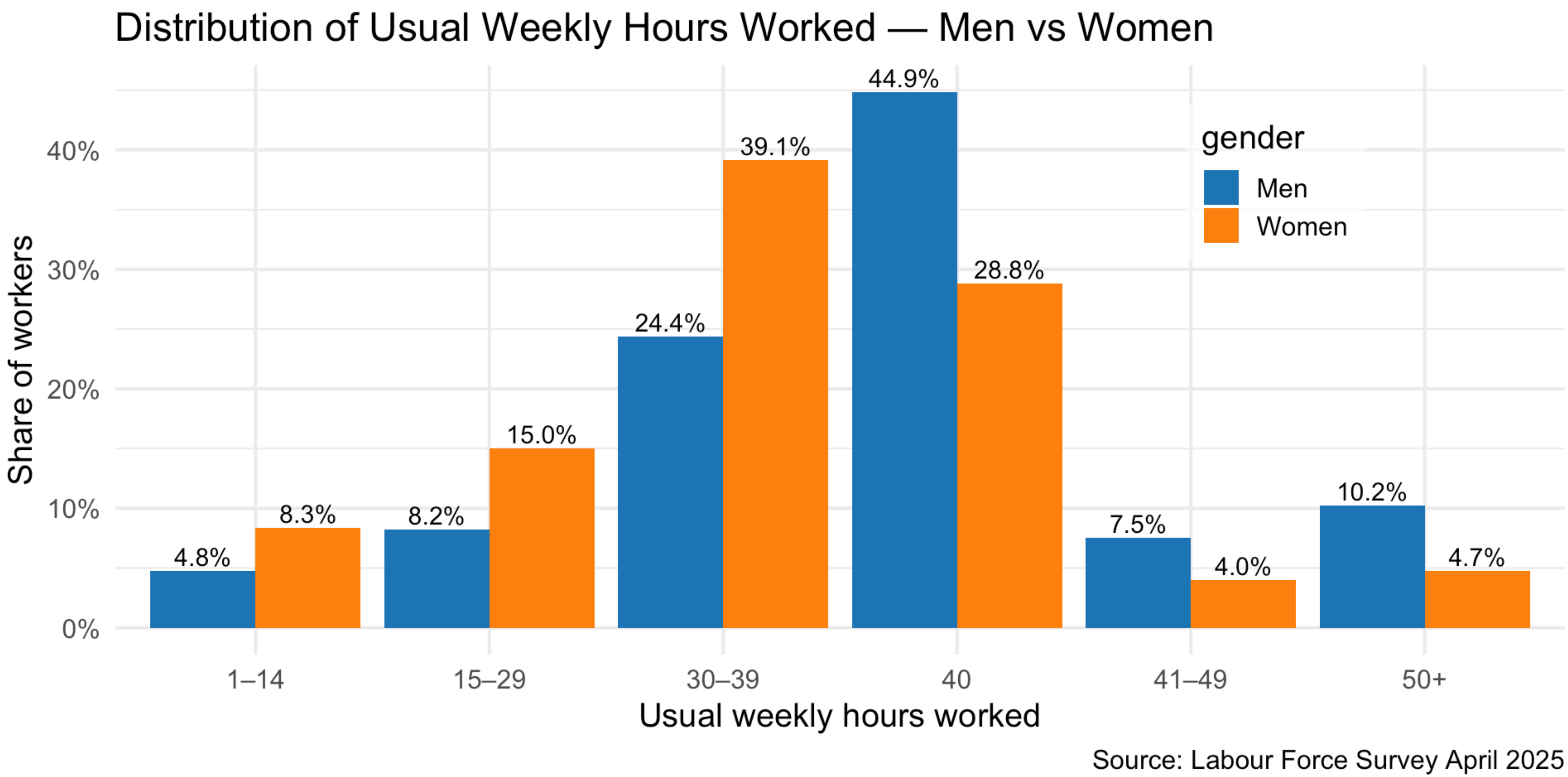
Labour Force Hierarchy



Hours Worked

- Individuals decide not only whether to work or not, but how intensively
- Some people work full time, some work part time
  - Work arrangements can be flexible
  - “Gig” jobs and remote work have increased that flexibility
- Recent data show
  - Almost half of men work full time hours
  - About 1/4 of women work full time hours

Hours Worked



Labour Supply Model

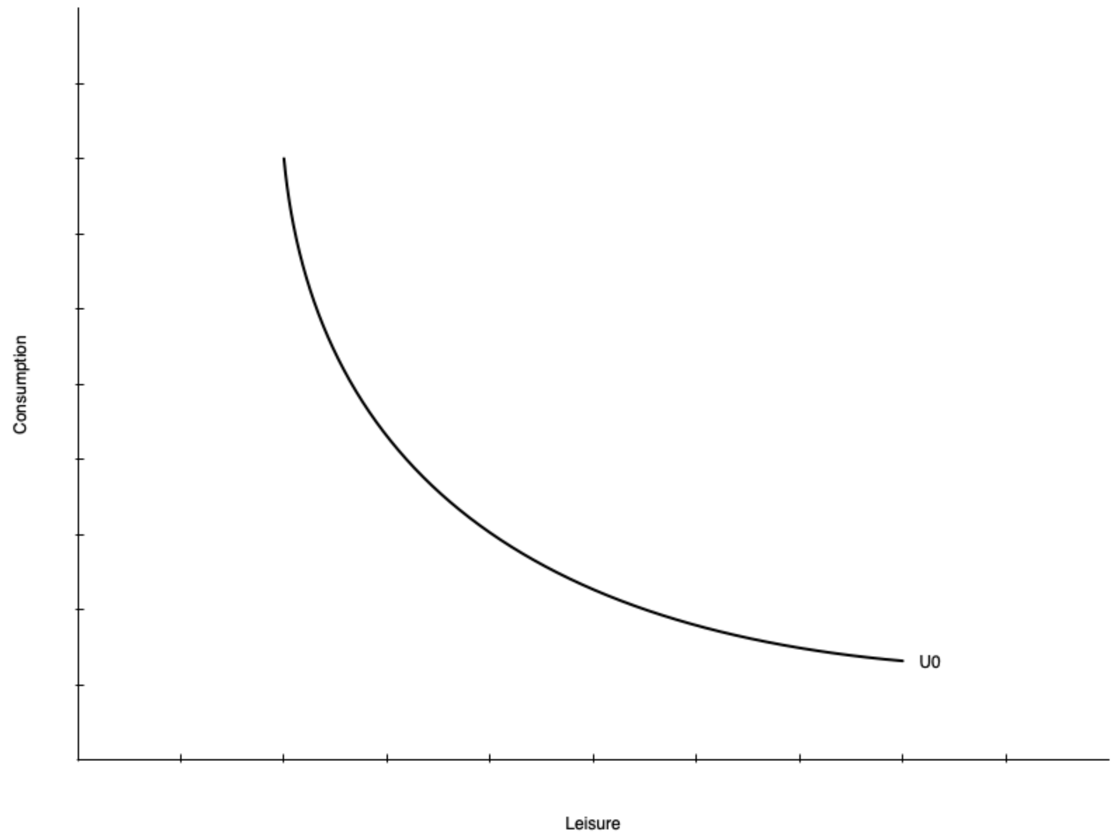
Introduction

- In this section we introduce a model we use throughout the course
- The model helps us understand how individuals make decisions about work
- Based on utility maximization as you saw in EC270
- Individual choose between two goods
    - Consumption of goods and services
    - Leisure
- Choices are constrained by time, wage rate, and non-labour income
- There are a few wrinkles that make this market unique

Preferences

- Individuals have preferences over consumption (C) and leisure (L)
  - People like consuming goods and services
  - Also like having non-work time
    - Includes fun activities but also non-work tasks like chores, education, etc.
- The tradeoffs people are willing to make between C and L are represented by indifference curves
  - At every point along the curve the individual is equally happy
  - Curves further from the origin represent higher levels of utility

Preferences

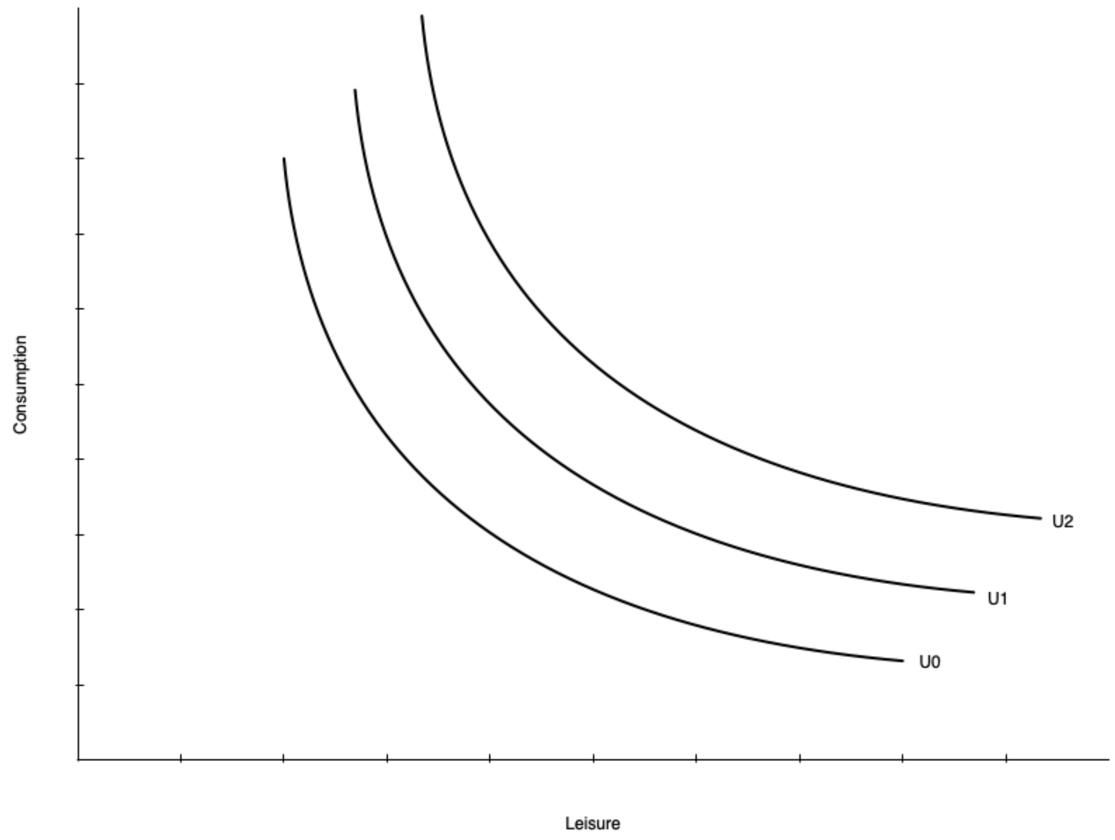


- On the left is a single indifference curve
- Any point represents equal utility (U_0)
- **Marginal Rate of Substitution (MRS)**: the rate at which a person is willing to trade consumption for leisure while remaining equally happy
- The MRS is the slope of the indifference curve at a point
- Individuals have diminishing MRS
  - Willing to give up less consumption for additional leisure the more leisure they already have
  - Slope gets shallower with more leisure

Preferences

- Indifference curves do not have to be smooth curves like the one shown
- Could be straight lines
  - Means the tradeoff between consumption and leisure is constant
  - Willingness to trade away leisure for consumption does not depend on leisure
- Could be “L” shaped
  - Consumption and leisure must be consumed in fixed proportions
- Mostly we will deal with smooth preferences

Preferences

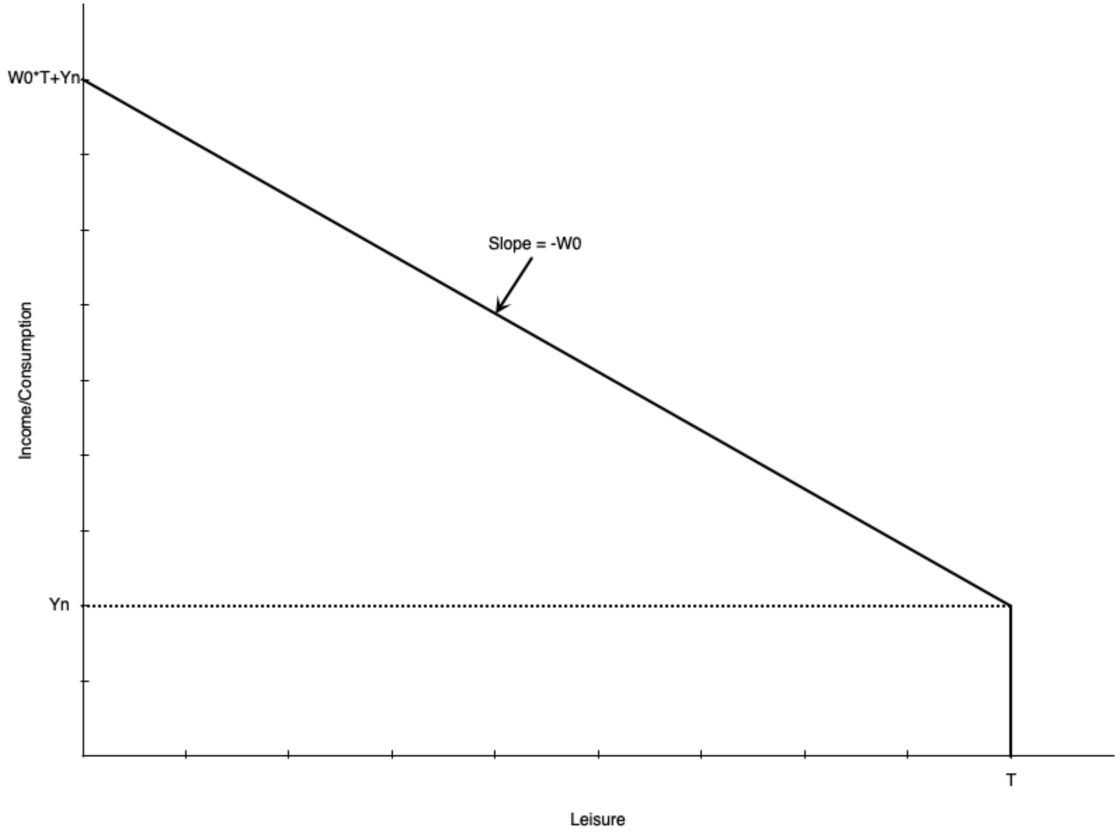


- Utility increases with curves further from the origin
  - $U_2 > U_1 > U_0$
-

Constraints

- Individuals do not choose consumption and leisure bundles on preference alone
- They are constrained by their budget
- The budget is determined by
  - The wage rate
  - Total number of hours available
  - Non-labour income
- Budget constraint is also called
  - Potential income constraint
  - Full income constraint

Constraints



- Graph shows a linear budget constraint
- Assume no saving, so consumption = income
- There are T total hours to allocate to work or leisure
- Individual earns Y_n non-labour income
  - Earned even with zero work hours (full leisure hours)
- Each hour worked gets wage of w
- Person working T hours (no leisure) earns $WT + Y_n$ income
  - Called "full income"

Constraints

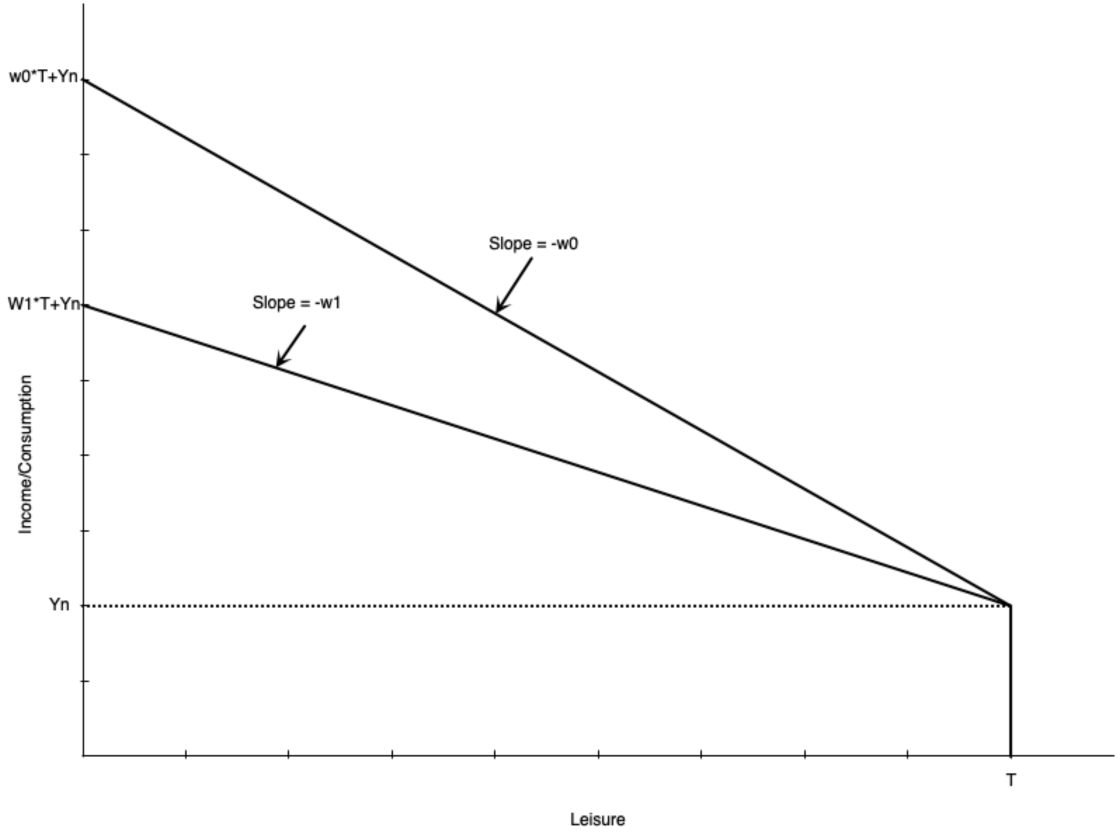
- Mathematically, the budget constraint is

$$C = w(T - L) + Y_n$$

- Consumption is equal to income from working $(w(T - L))$ plus non-labour income (Y_n)

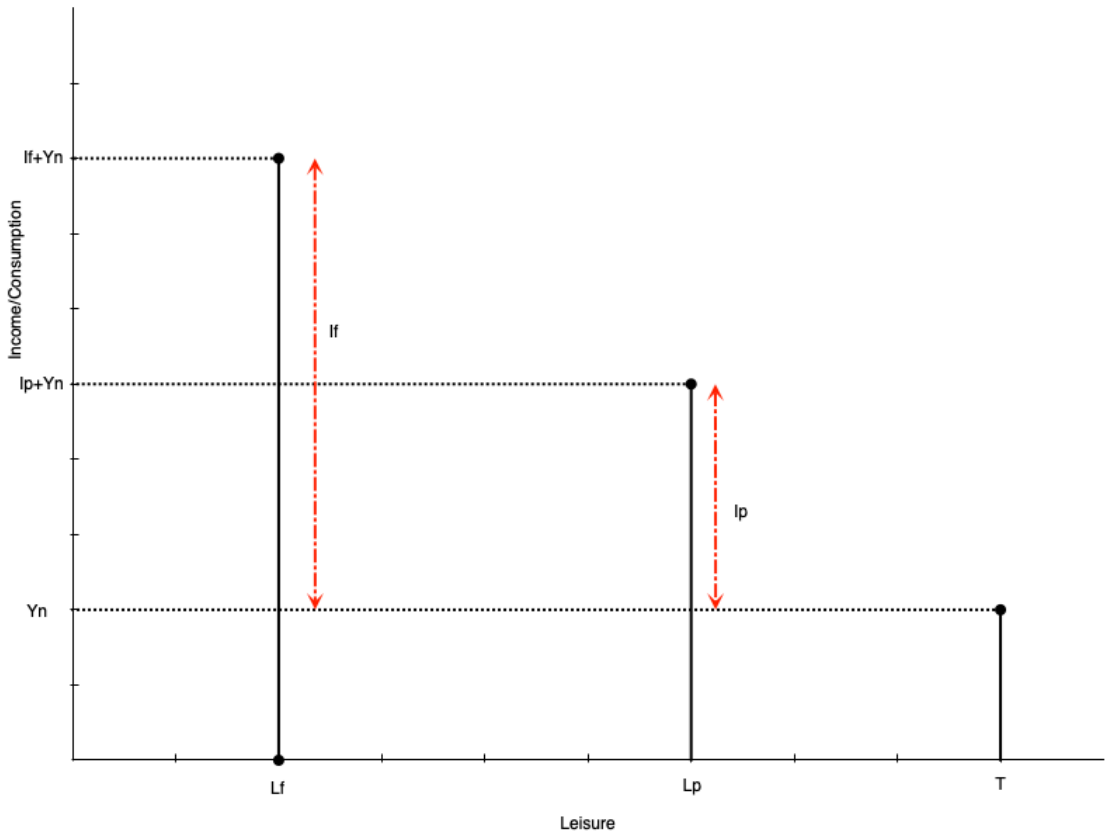
- The slope of the budget constraint is $-w$
  - The opportunity cost of leisure is the wage rate
  - Each hour of leisure means one less hour worked, which means w less income
- Someone who works zero hours has T leisure hours and Y_n consumption
- Someone who works T hours has zero leisure hours and $wT + Y_n$ consumption

Constraints



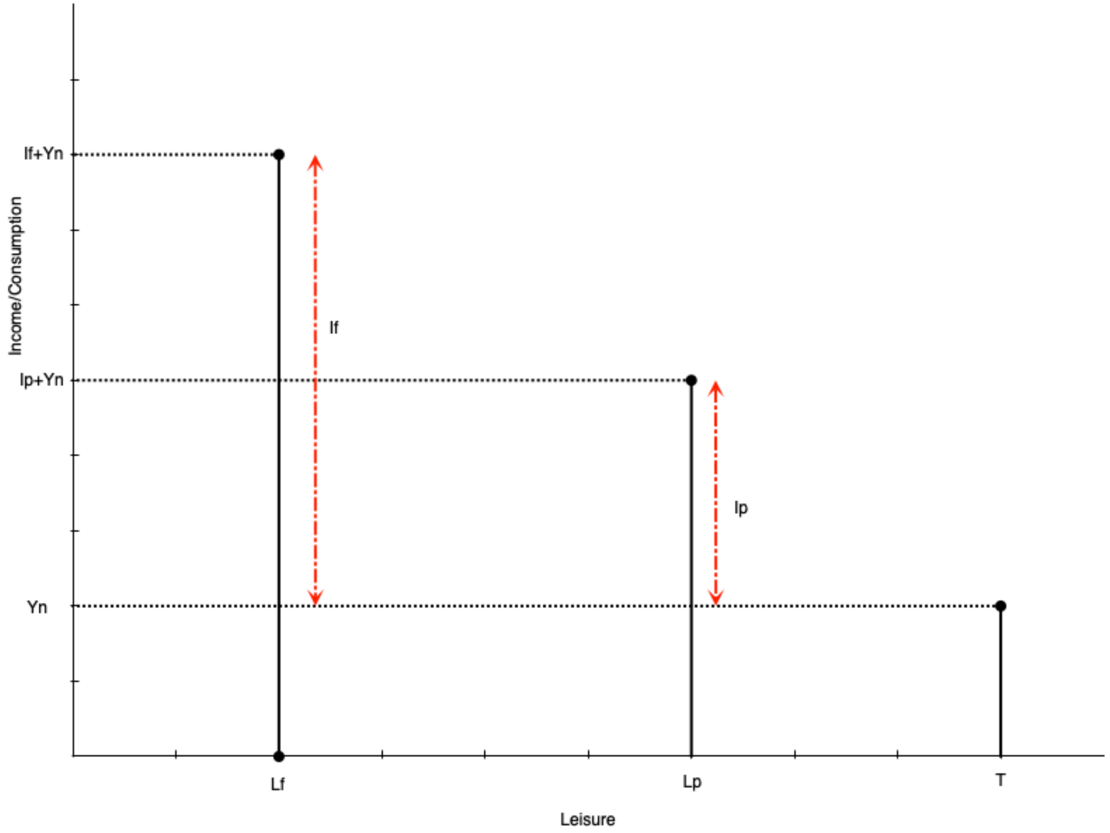
- Slope is the negative of the wage rate w
- A lower wage means shallower slope
  - If $w_1 < w_0$, the budget line swivels down
  - They intersect at leisure = T because there are no work hours
- Lower wage also means full income is lower
  - Working all hours leads to less total labour earnings

Constraints



- Often people cannot adjust their work hours freely
- They make only be able to work full time or part time
- Budget constraint is no longer a line, but a set of points
- Suppose an individual chooses part time work
    - Work hours are $h_p = T - L_p$
    - Labour earnings are $I_p = w * h_p$
    - Total consumption (and income) is $C_p = I_p + Y_n$

Constraints

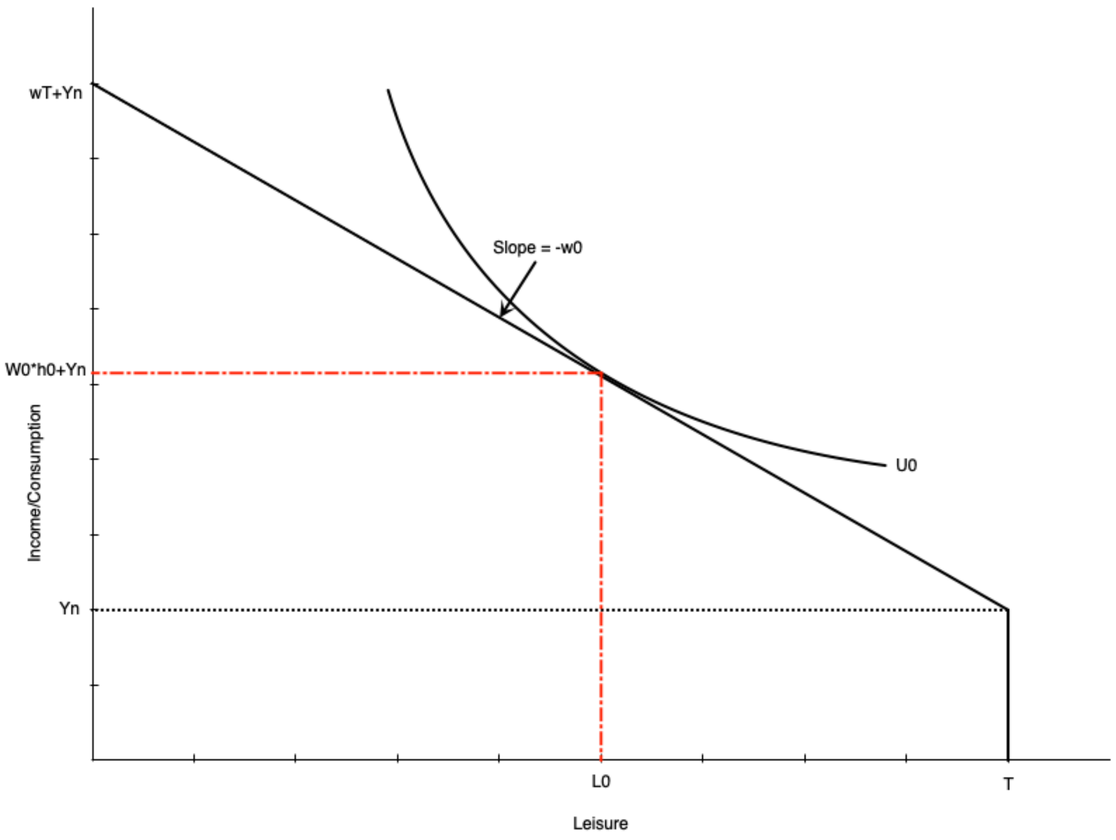


- A person working full time hours
  - Work hours are $h_f = T - L_f$
  - Labour earnings are $I_f = w * h_f$
  - Total consumption (and income) is $C_f = I_f + Y_n$
- In theory wage could also be different for full time work

Consumer Optimum

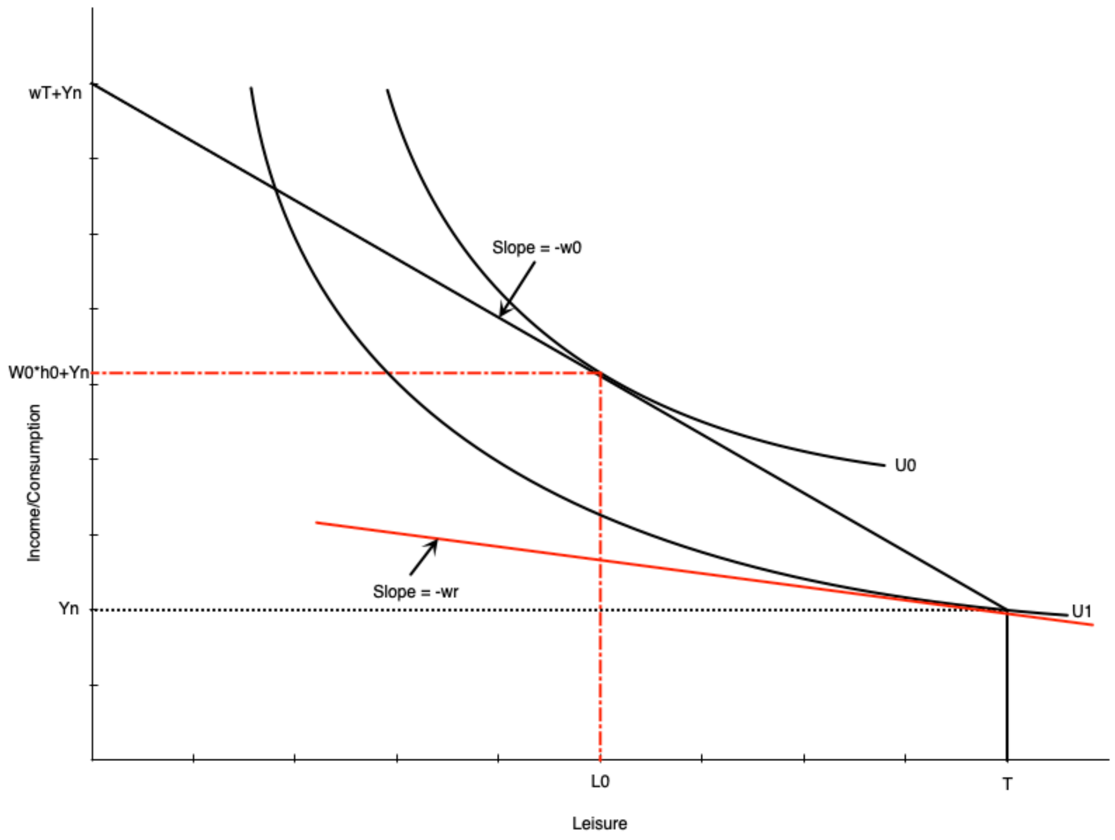
- Work hours are chosen using the combination of preferences and constraints
- The consumer optimum is the point that maximizes utility subject to the budget constraint
- The optimum occurs where the budget constraint is tangent to an indifference curve
- There are two possibilities for the optimum
  - Non-participation: optimum occurs at the end of the budget constraint (full leisure)
  - Participation: optimum occurs at an interior point (some leisure, some work)
- Model also defines the **reservation wage** -Lowest wage rate at which a person will choose to work

Consumer Optimum



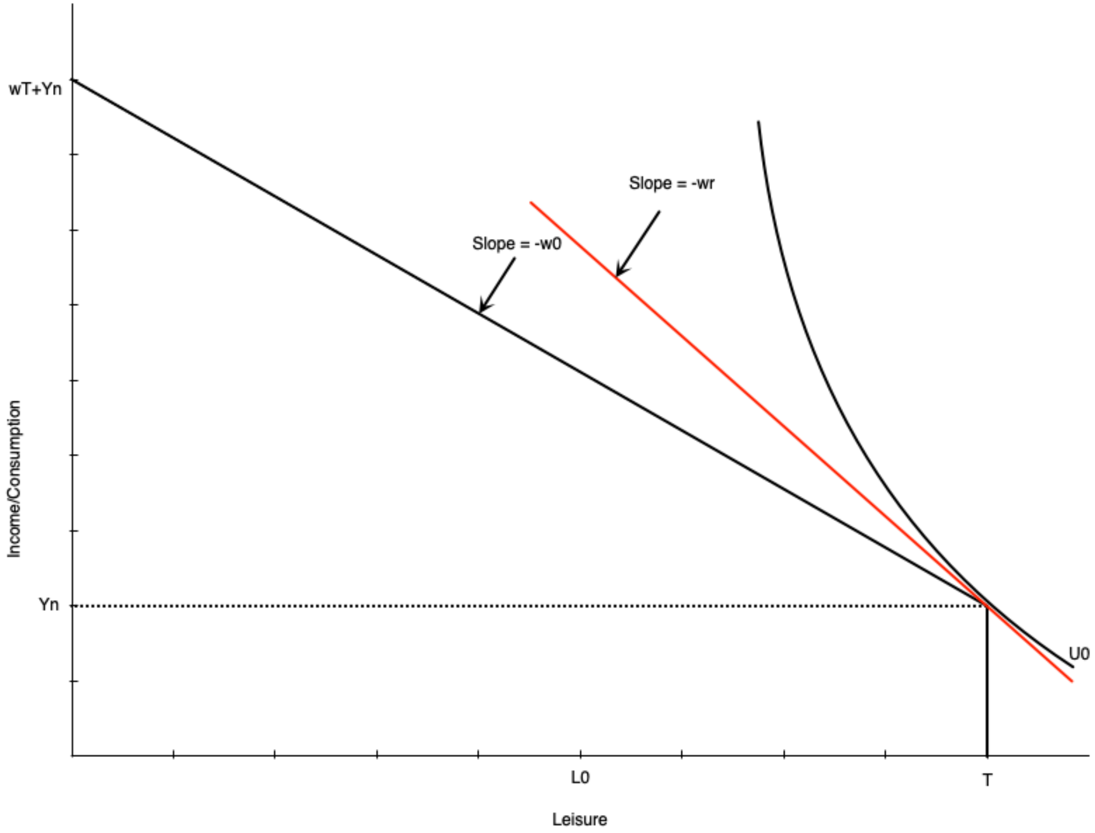
- Graph shows an optimum with participation
- Optimum is where indifference curve is tangent to budget constraint
- At the optimum, MRS = wage rate
  - Willingness to trade leisure for consumption (MRS) equals ability to trade them (wage)
- Optimal work hours are $h_0 = T - L_0$
- Optimal consumption is $C_0 = W_0 h_0 + Y_n$

Consumer Optimum



- The reservation wage is the lowest wage at which a person will choose to work
- It occurs where the budget constraint is tangent to the indifference curve at full leisure
  - At that point, MRS = reservation wage
- If the wage is any lower, the person will choose not to work
- If it is any higher, they will work positive hours
- The red line shows the hypothetical reservation wage
  - In this case, it is w_r

Consumer Optimum



- Graph to the left shows a non-participation optimum
- The slope of the indifference curve is steeper than the budget constraint at full leisure
  - The extra consumption you want from one less hour of leisure (MRS) is more than you can get from working an hour (wage)
- So your optimal choice is to not work at all
- Indifference curve touches the budget constraint at the kink
- Notice the wage w_0 is lower than the reservation wage w_r

Comparative Statics

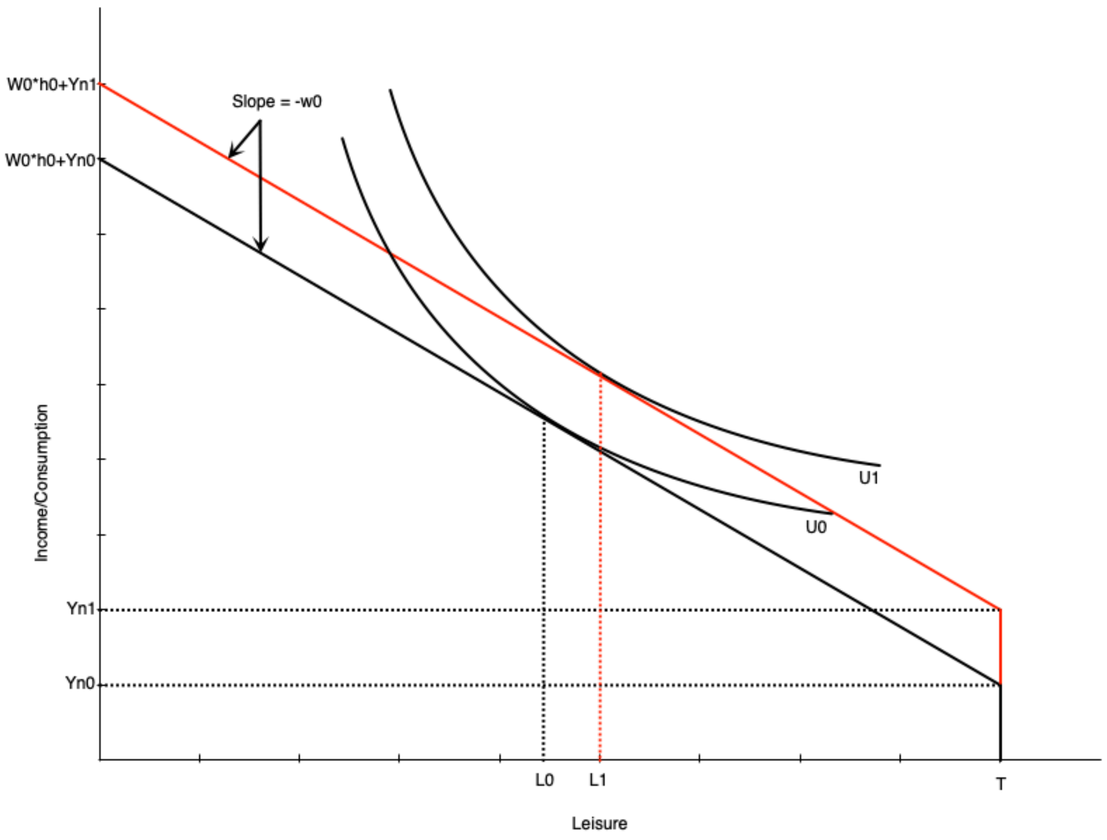
Introduction

- The model can be used to predict how changes in the environment affect work decisions
- Changes could be to the wage or non-labour income
- Often these arise because of policy changes
  - Introduction or increase in welfare payment
  - Wage subsidy
  - Tax credits
  - Employment insurance
  - Child care
- We can examine these change with our model

Increase in Non-Labour Income

- Suppose that individuals receive an increase in non-labour income
  - Could be from a government transfer program
- How does this affect optimal work hours?
- Depends on whether leisure is a normal or inferior good
  - Normal good: leisure increases when non-labour income increases
  - Inferior good: leisure decreases when non-labour income increases
- Generally economists think of leisure as normal

Increase in Non-Labour Income

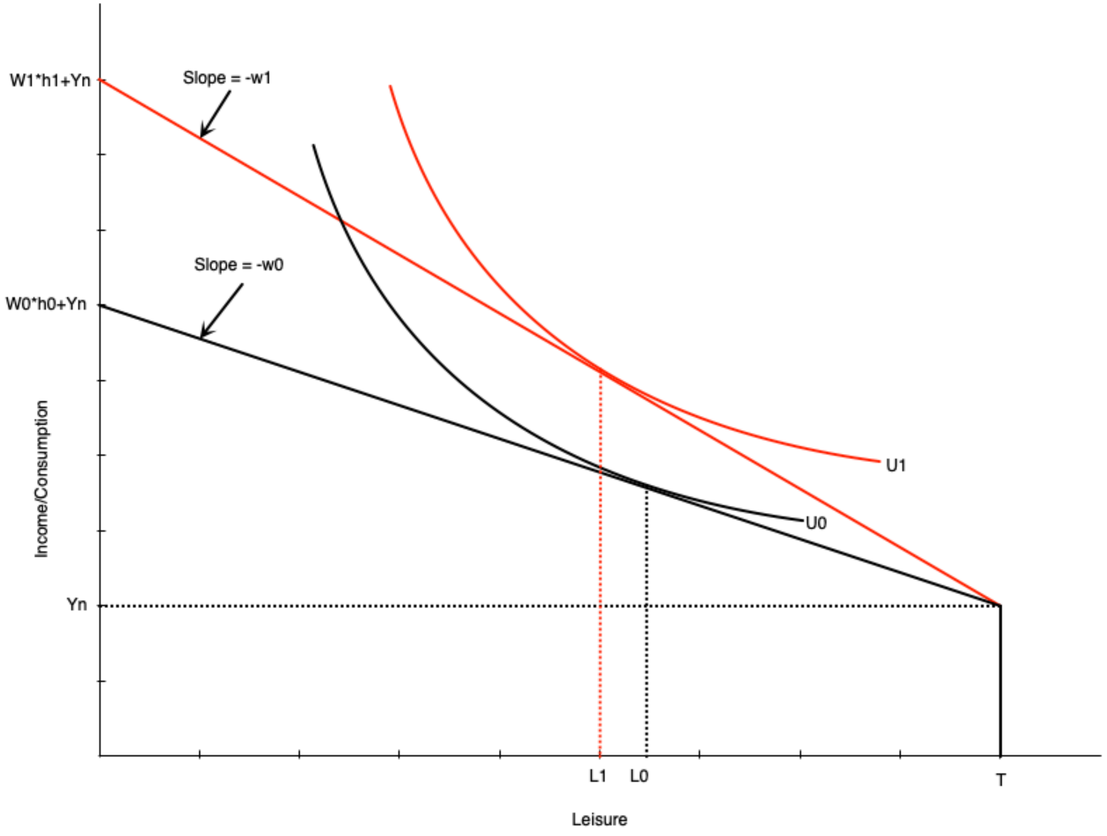


- Graph shows an increase in non-labour income from Y_{n0} to Y_{n1}
- Budget line shifts up vertically but parallel
  - No change in slope because wage is the same

-If leisure is a normal good, optimal leisure increases from L_0 to L_1

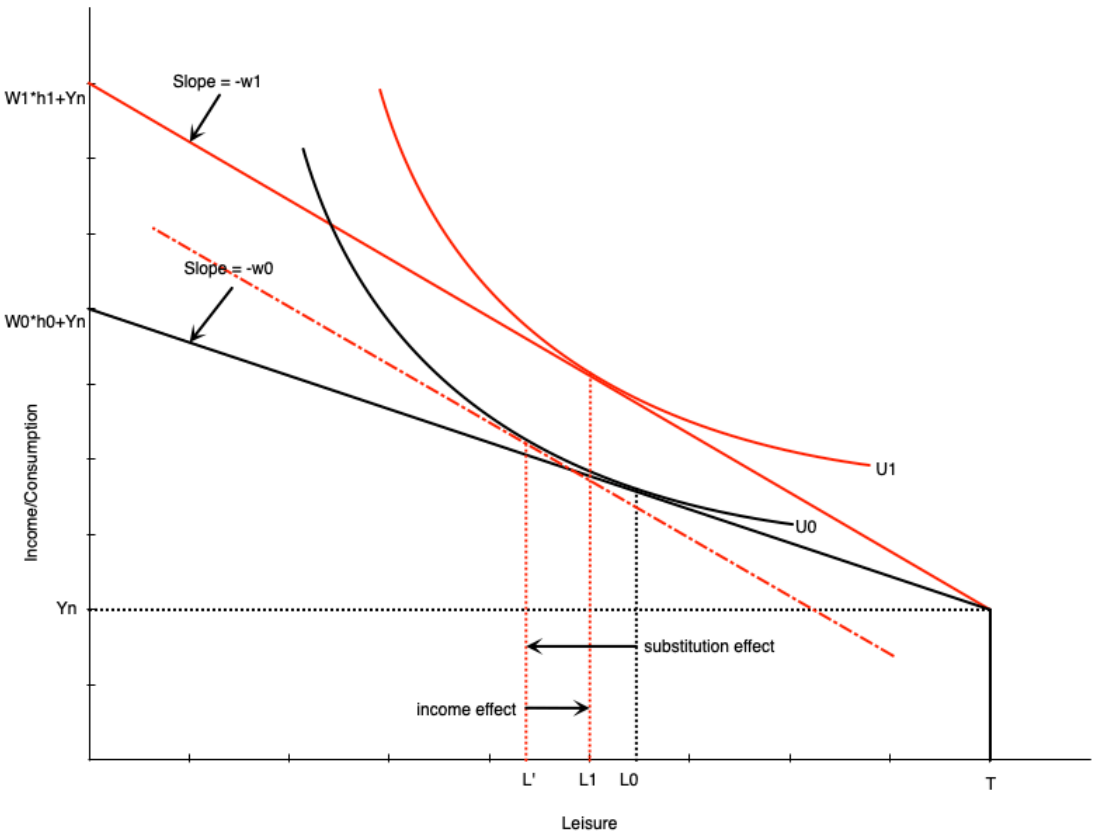
- Means work hours decrease by the same amount
- Consumption also increases
- This illustrates a pure **income effect**
  - Increases in income increase demand for normal goods

Increase in the Wage



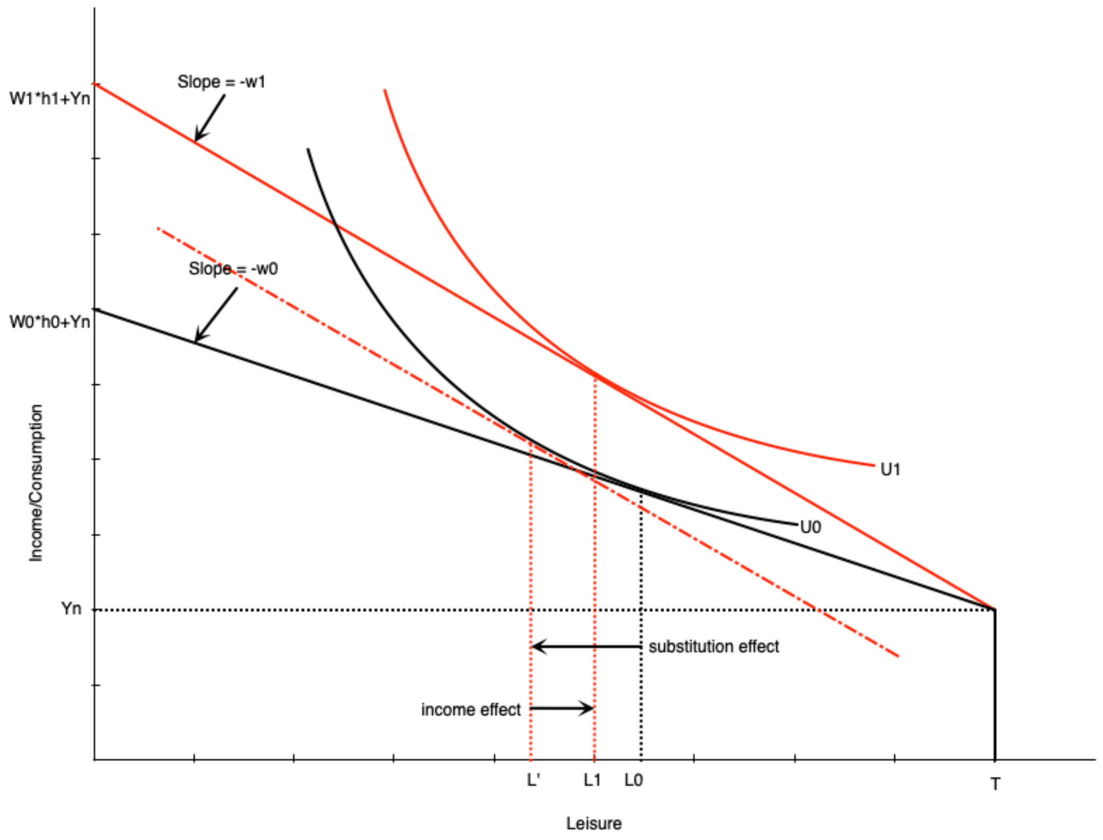
- If the wage changes, the budget constraint pivots
- In graph to the left, the wage increases from w_0 to w_1
- The slope becomes steeper
- Full income also increases because working all hours earns more
- Graph is drawn such that leisure falls from L_0 to L_1 at the new wage rate
- There are two effects underlying this change
  - Substitution effect
  - Income effect

Increase in the Wage



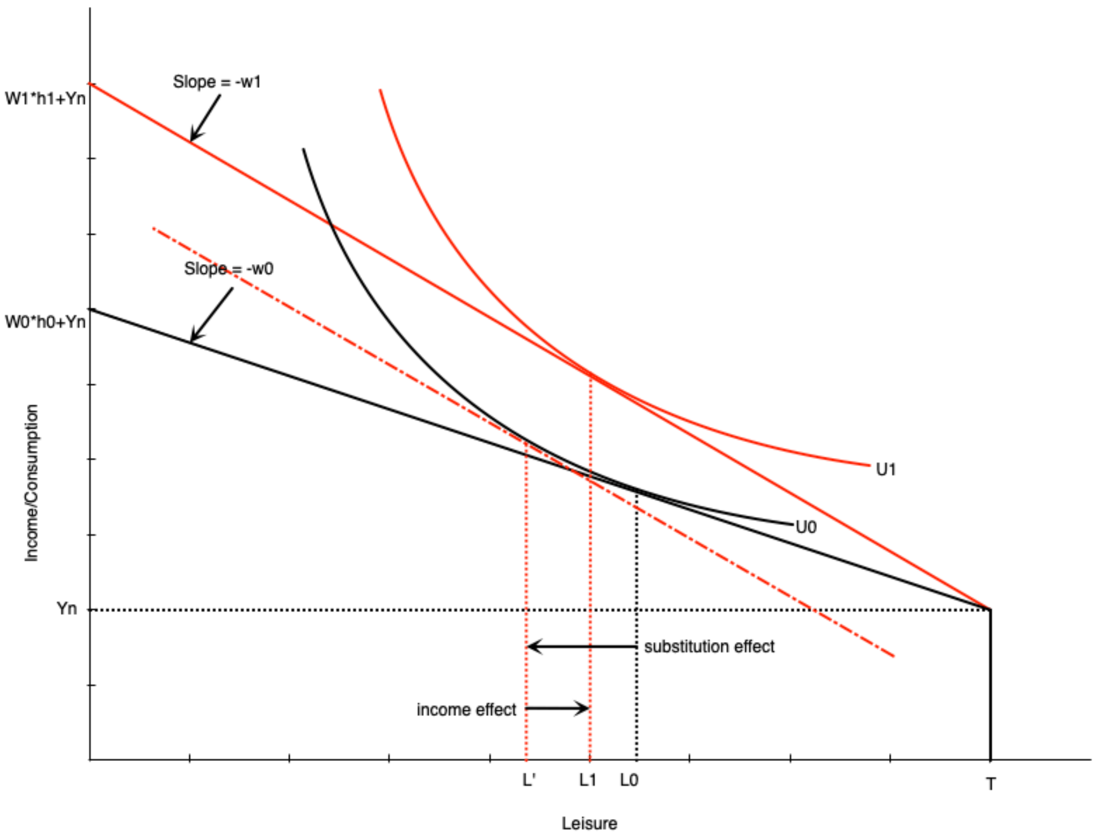
- **Substitution Effect**: change in leisure when wage changes but utility is held constant
  - When wage increases, leisure becomes more expensive
  - People substitute away from leisure and towards consumption
  - To measure this, draw a hypothetical budget line with the new slope but tangent to the original indifference curve
  - Difference in leisure measures substitution effect

Increase in the Wage



- **Income Effect**: change in leisure when income changes but wage is held constant
  - When wage increases, workers become richer at same hours of work
  - Want to consume more of all normal goods
  - To measure this, move from hypothetical budget line to new budget line
  - Difference in leisure measures income effect

Increase in the Wage



- The income and substitution effects move in opposite directions
- For a wage increase
  - Substitution effect encourages more work (less leisure)
  - Income effect encourages less work (more leisure)
- The overall effect on work hours depends on which effect is stronger
- Graph on the left shows a stronger substition effect